

Empirical M_w – M_L , m_b , and M_s Conversions in Western China

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Abstract Local magnitude (M_L), body-wave magnitude (m_b), and surface-wave magnitude (M_s), which are saturated at certain values and may lead to an incorrect energy estimation of a large earthquake, are largely used in quantifying the size of an earthquake in western China. Based on the catalog from the China Earthquake Data Center (CEDC) and moment magnitude (M_w) provided by the Global Centroid Moment Tensor (Global CMT) Project, we test the general orthogonal regression (GOR) and the ordinary least-squares (OLS) methods in M_w – M_L , M_s , and m_b conversions for earthquakes in three different tectonic structures of western China. For M_L and m_b , linear trends vary according to tectonic structures, which implies different body-wave attenuations in the three structures. For M_s , the results are similar, whether using the GOR or the OLS method, but slopes of regression lines are not close to 1, which indicates routine misestimates of M_s in western China. The GOR slopes are uniformly larger than the OLS slopes in all magnitude conversions. Standard deviations are between 0.06 and 0.13 for the GOR method but are between 0.12 and 0.25 for the OLS. Thus, the GOR is found to be superior to the OLS method and its use is recommended. Conversions of different magnitudes to M_w not only imply different patterns of seismic-wave attenuations but will also benefit immediate assessment of seismic damage after occurrence of a destructive earthquake in the future.

Online Material: Earthquake catalog and tables showing differences between magnitude estimates for M_w , M_L , m_b , and M_s .

Introduction

Since the first magnitudes scale, so-called local magnitude (M_L ; Richter, 1935; Hutton and Boore, 1987), was developed to quantify earthquake size, many scales were created to estimate earthquake magnitudes. The M_L is now widely used in different tectonic environments (Ebel, 1982; Bakun and Joyner, 1984; Alsaker *et al.*, 1991; Kim, 1998). Most modern methods for estimating common earthquake magnitude (e.g., surface-wave magnitude M_s and body-wave magnitude m_b) were designed to be consistent with the M_L . In China, a general form for estimating M_L is

$$M_L = \log A + R(\Delta), \quad (1)$$

in which A and $R(\Delta)$ (Table S1, available in the electronic supplement to this article) are the maximum average amplitude in horizontal components (in μm) and a correct term of regional attenuation, respectively. The M_L is usually not used for earthquakes with $M_L \geq 7$ or epicentral distance ≥ 1000 km in China. Instead, m_b and M_s are often used for determining magnitudes of earthquakes for which M_L cannot be accurately estimated.

For earthquakes occurring in western China, m_b and M_s are largely used in size estimation, due to distribution of seismic stations. The m_b is measured from P wavetrain and is defined as

$$m_b = \log(A/T) + Q(h, \Delta), \quad (2)$$

in which A is the maximum amplitude of P wavechain in micrometers, T is the wave period in seconds, usually ~ 1 s, and $Q(h, \Delta)$ (Table S2) is an empirical term depending on the focal depth (h) and epicentral distance (Δ) in degrees. The M_s is measured using the largest amplitude of the surface waves and is usually written as

$$M_s = \log(A/T) + 1.66 \log \Delta + 3.5, \quad (3)$$

in which A is the maximum horizontal amplitude in micrometers with a certain period T , and Δ is the epicentral distance in degrees. For large distant earthquakes, the m_b and M_s can be directly measured from seismograms without specific data analysis.

It has been known that there is no significant difference between different scales for earthquakes with magnitude ≤ 4.5 in China (Liu *et al.*, 2007). For events with magnitude > 4.5 , the discrepancies would become greater. Besides, magnitudes reported by China government agencies and by the National Earthquake Information Center (NEIC) of the U.S. Geological Survey (USGS) have consistent differences for large earthquakes that have occurred in China (Liu *et al.*, 2005, 2006). For

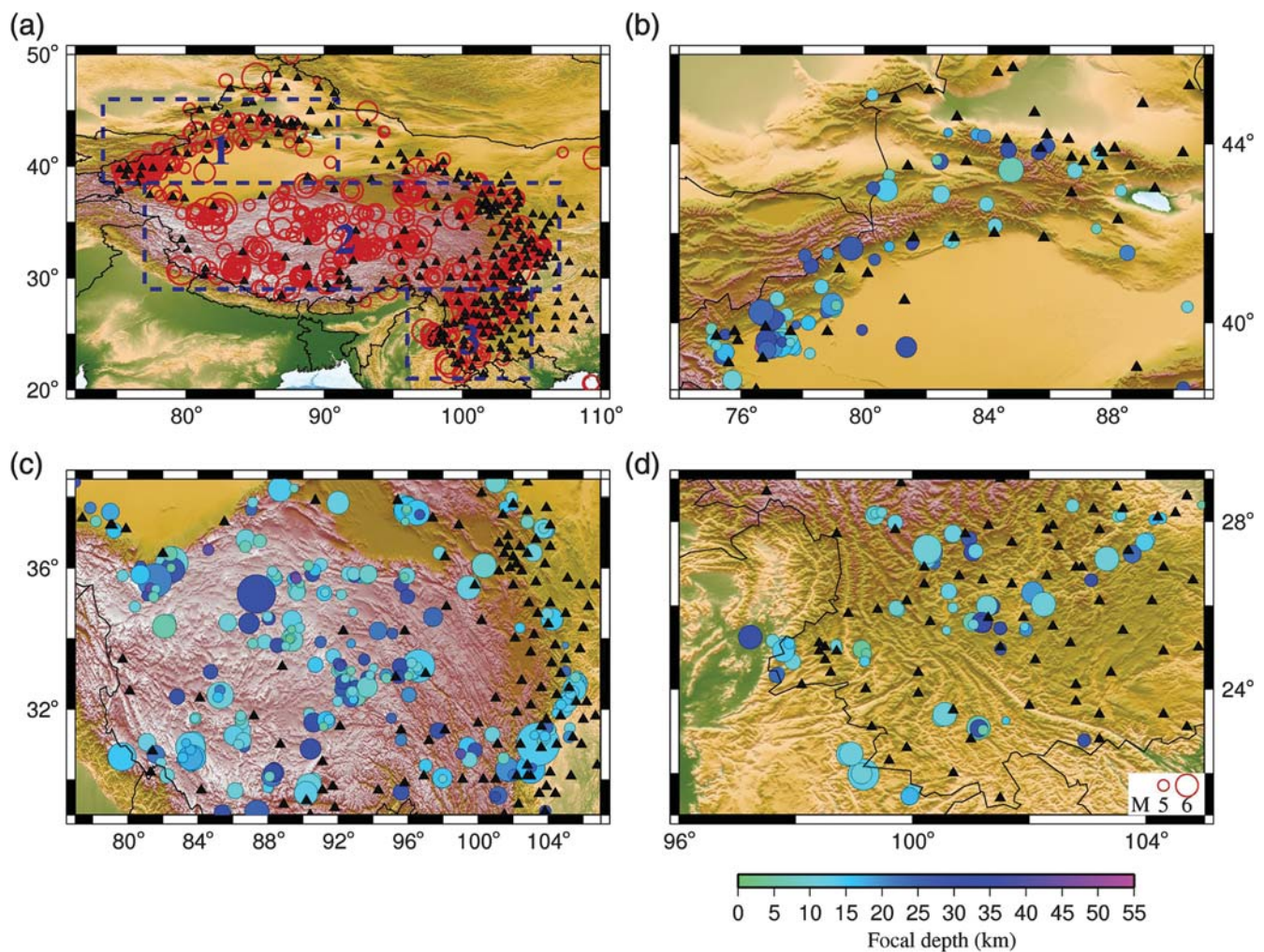


Figure 1. (a) Epicenters of large earthquakes with $M_w > 4.5$ (red open circles) in western China during 1990 and 2014. Study areas are bounded by blue dashed lines and shown in (b)–(d). (b) Zone 1: longitude between 74° E and 91° E and latitude between 38.5° N and 46° N. (c) Zone 2: longitude between 77° E and 107° E and latitude between 29° N and 38.5° N. (d) Zone 3: longitude between 96° E and 105° E and latitude between 21° N and 29° N. Black triangles represent seismic stations. The earthquake magnitudes are shown in (b)–(d) with corresponding sizes.

example, M_s determined by Institute of Geophysics China Earthquake Administration are larger by 0.2 than those determined by the NEIC for earthquakes with $M_s \geq 7.0$.

Moment magnitude (M_w) is directly connected to earthquake source processes, does not saturate, and thus provides the most robust estimate of the magnitude of large earthquakes. The M_w is derived from the seismic moment using seismic wave with much longer periods (Kanamori, 1977; Hank and Kanamori, 1979). Moment tensors for earthquakes with magnitude larger than 5 are now routinely estimated by the Global Centroid Moment Tensor (Global CMT) Project operated by Columbia University (Dziewonski *et al.*, 1981; Ekström *et al.*, 2012).

In China, most of the large events occurred in the west region from 1990–2014 (Fig. 1). So far, part of earthquakes that occurred in the region will not be given M_L due to distribution of seismic stations (Fig. 1). The m_b and M_s are usually calculated for moderate-to-large earthquakes in this region. Nevertheless, the equations for estimating M_L , m_b , and M_s are totally empiri-

cal, and these magnitudes saturate beyond certain values so that they have no direct connection to the physics of earthquakes. Therefore, it is necessary to calibrate these magnitude scales with an unsaturated one such as M_w , which is directly connected to the energy release of an earthquake. More than 480 moment tensor solutions for events with $M_w > 4.5$ in the region have been calculated by the Global CMT Project in 1990–2014. It allows empirical M_w – M_L , M_w – M_s , and M_w – m_b conversions and helps to give a consistent scale. We propose that the conversions may offer quick estimation of the energy release of a large earthquake and rapid assessment of seismic hazard in western China in the future.

Tectonic Setting of Western China

Western China contains three major tectonic units, the Tianshan earthquake belt (TEB, zone 1), the Tibetan plateau (TP, zone 2), and the southern segment of the north–south seismic belt (NSSB, zone 3). Xu *et al.* (2000) reported that the

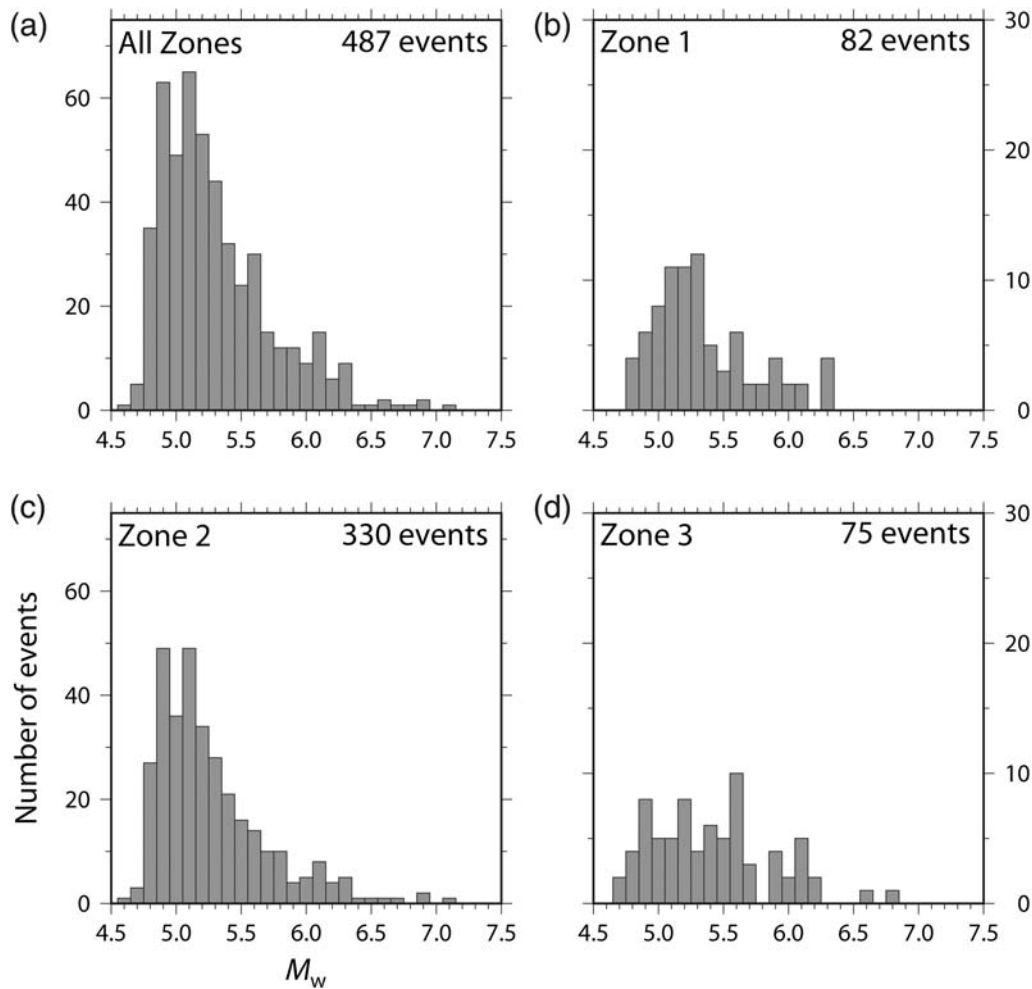


Figure 2. Histograms showing the number of events and M_w distribution for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. Numbers of events used in this study are also shown in each figure.

crustal thickness under the TEB is more than 50 km, and that there is a low-velocity zone (LVZ) in the middle crust. Previous studies indicated that the TEB is oriented in the northeast–east–southwest–west direction and that large earthquakes mainly occurred in the mid–lower crust beneath the western TEB (Qu *et al.*, 2008; Wang and Wang, 2010). The TP located south of the TEB is a region with high tectonic deformation and seismicity. This results from the collision between the Eurasian plate and the Indian plate. Historical records show that there have been more than 100 earthquakes with magnitude > 7 in this region since 1900. The seismicity in TP shows a high risk of seismic hazard (Deng *et al.*, 2014). Based on body- and surface-wave tomography, there are several LVZs that may reflect complex geodynamic processes beneath the TP (Ding *et al.*, 2001; Li *et al.*, 2009). The southern segment of NSSB is located at the southeastern borderland of the TP. There are several strike-slip faults and deformation belts in the segment of NSSB (Zhang *et al.*, 2003). Previous studies based on surface-wave and ambient-noise tomography also showed existence of LVZs beneath the

western segment of NSSB (Yao *et al.*, 2008; Huang *et al.*, 2013). These LVZs may connect with the LVZs of the TP and implied lateral flow of lower crustal material.

Analyses Procedure

We first selected earthquakes that occurred during 1990–2014 in western China from the Global CMT catalog and then found the M_L , m_b , and M_s of these events from a catalog reported by the China Earthquake Data Center (© Table S1) that collects data from more than 700 seismic stations (see Data and Resources). To better obtain empirical conversion, we limited our analyses to crustal earthquakes with focal depths < 55 km. Next, we divided the study region into three zones according to different tectonic settings (Fig. 1).

The M_w of the events range from 4.5 to 7.9 (© Table S3). Our data set provided by the CEDC includes three events with $M_w \geq 7.5$: the 1997 Tibet M_w 7.5, the 2001 M_w 7.8 Kunlun, and the 2008 M_w 7.9 Wenchuan earthquakes. Their magnitudes may be saturated in M_L , m_b , and M_s scales. Thus, we only focus on events with

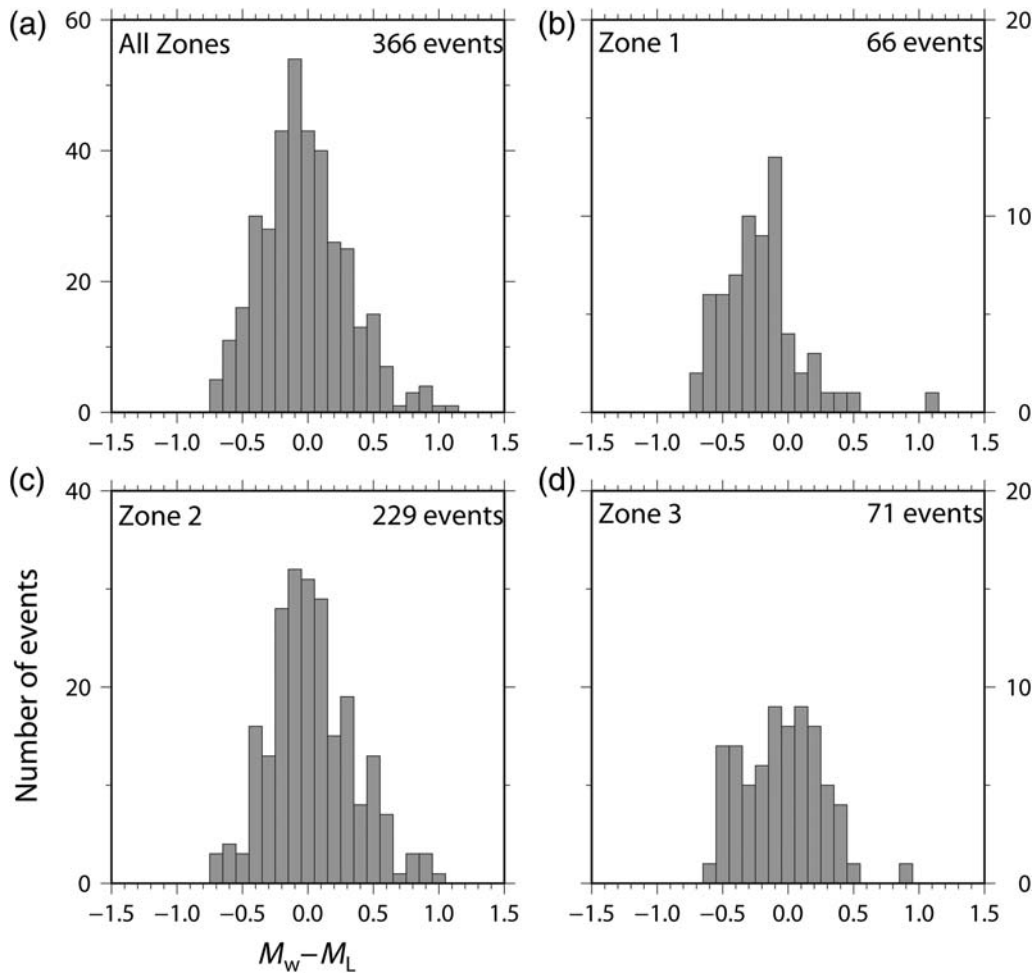


Figure 3. Histograms of the M_w-M_L differences for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. The differences are mainly between -0.5 and 0.5 . Numbers of events are also shown in each figure.

$4.5 < M_w < 7.5$ in the regression analysis. A total of 366, 487, and 487 events were used in this study for zones 1, 2, and 3, respectively (Fig. 2). We determined the best-fit linear relationship between different magnitudes and M_w . The linear regression models are separately computed by the ordinary least-squares (OLS) and the general orthogonal regression (GOR) methods (Castellaro *et al.*, 2006; Castellaro and Bormann, 2007; Gasperini *et al.*, 2015) shown as

$$M_w = aM + b, \quad (4)$$

in which a and b are constants to be determined from the regression analysis. M represents M_L , m_b , or M_s . Because the error variance ratio (η) for different magnitudes cannot be estimated, we followed the procedure by Castellaro *et al.* (2006) and arbitrarily set it to 1 in the GOR.

Results

M_w-M_L Conversion

Figure 3 shows M_w-M_L differences of 366 events having M_L values in the CEDC catalog for the study regions.

Basically, the M_w-M_L differences are between -0.5 and 0.5 (Fig. 3a). Among three zones, the distribution of M_w-M_L differences of zone 1 (Fig. 3b) is distinct from the other two zones (Fig. 3c,d). In zone 1, the M_w-M_L differences are between -0.6 and 0.0 , which indicate the M_L values were largely overestimated. For zones 2 and 3, the differences basically concentrate around 0.

The results calculated by the GOR and OLS methods for M_w-M_L are shown in Figure 4. Most events have M_L of $4.5-6.5$, and the largest value is 7.1. The slopes of regressed line from the GOR (solid red) and the OLS (solid blue) methods are 0.71 and 0.61 for the whole study area, respectively (Fig. 4a). In zone 1, the slopes derived from both the GOR and OLS (solid blue) methods are lower than the slopes in zones 2 and 3, mainly because M_L is larger than M_w for all of the earthquakes with $M_L \geq 5.5$ (Fig. 4b-d). The line slopes calculated by the GOR method are ~ 0.1 higher than those from the OLS method in average. The GOR slopes vary from 0.66 to 0.75, with a standard deviation of ~ 0.13 , significantly lower than those obtained by the OLS method (Fig. 4b-d).

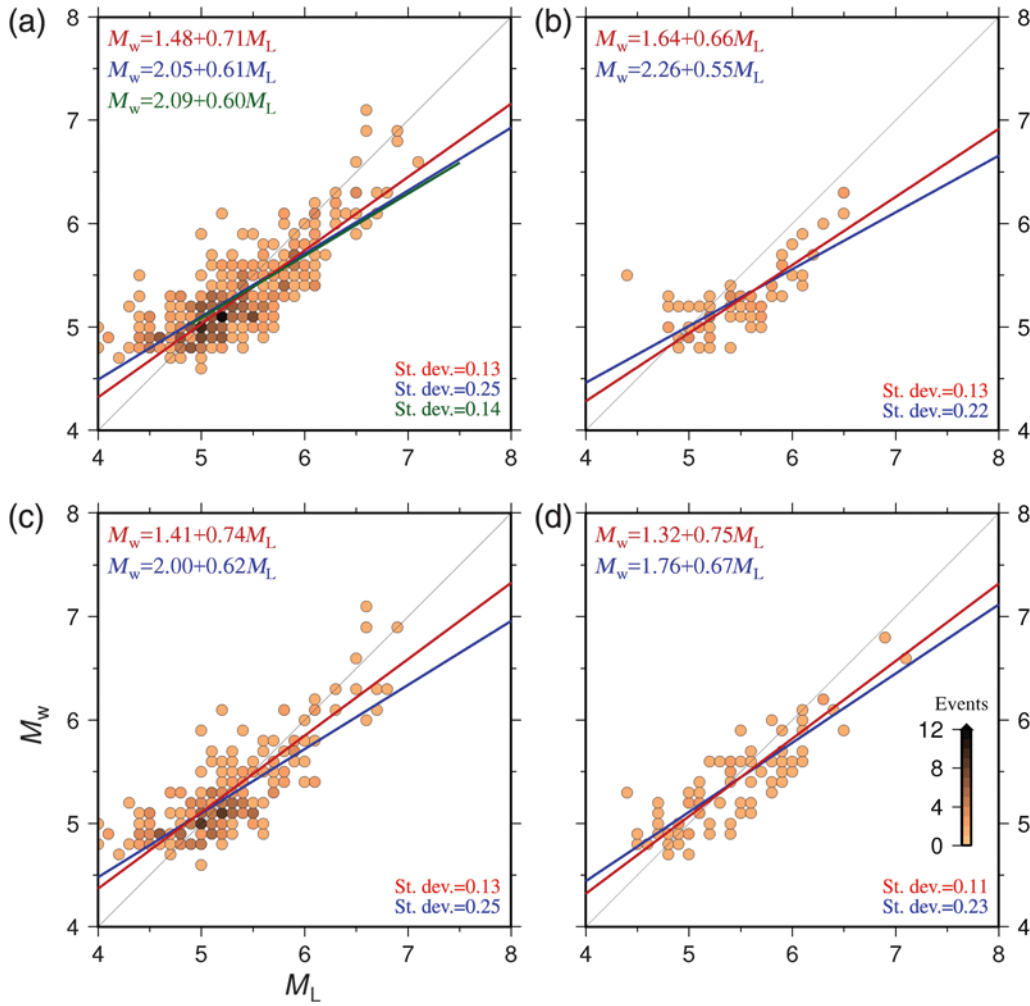


Figure 4. M_w – M_L calibrations for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. The red and blue lines represent regression results calculated by the general orthogonal regression (GOR) and ordinary least-squares (OLS) methods, respectively. The green line, which starts at M_L 5.0 and ends at M_L 7.5, indicates the GOR regression of using 446 events whose $M_w \geq 4.9$. Standard deviations (st. dev.) with different colors correspond to regression lines. The colored circles represent occurrence times of events. One black circle represents the times over 12. The thin gray line represents a 1:1 relationship between M_w and M_L . The solid red line is a best-fit line from regression analyses. One standard deviation is shown by the blue dashed line.

M_w – m_b Conversion

We have 487 earthquakes with both M_w and m_b in the range $4.3 \leq m_b \leq 6.3$ (Table S1). The M_w – m_b differences are shown in Figure 5 for the study regions. Most of the differences are larger than 0.1, regardless of regions, and the largest value of the differences is 1.1. The catalog indicates that most m_b might be routinely underestimated in western China.

Figure 6 shows M_w – m_b regression results. On average, the slopes of the regressed line from the GOR and the OLS methods are 1.16 and 0.95 for western China, respectively (Fig. 6a). The line slopes derived from the GOR for zones 1 and 3 (1.23 and 1.28, respectively) are similar but larger than the slope for zone 2 (1.12; Fig. 6b–d). Distinguished from the GOR results, the OLS slopes for zones 1 and 2 are 0.94 and 0.92, respectively, but smaller than the slope of 1.05 for zone 3. Compared to the OLS regressions, the GOR results

have larger divergences from 1:1 relation but have smaller standard deviations of ~ 0.1 . Plots of M_w versus m_b imply that m_b saturates at ~ 6.2 .

M_w – M_s Conversion

A total of 487 events with M_s between 4.3 and 7.5 are listed in the CEDC catalog, and a large part of the M_w – M_s differences are between -0.4 and 0.4 (Fig. 7). For zones 1 and 2, M_w – M_s difference distributions (Fig. 7b,c) are similar to that of all zones (Fig. 7a). Nevertheless, the M_w – M_s differences for zone 3 are mainly between -0.6 and 0.2 (Fig. 7d). The regression results for M_s to M_w are shown in Figure 8. It is clear that the M_s presents a good linear correlation with M_w whether using the GOR or the OLS method. For each zone, the slopes of best-fit lines are similar and close to 0.68, which lies below the 1:1 relationship. The slopes of best-fit lines are close to 0.68, both in the GOR and the OLS analyses.

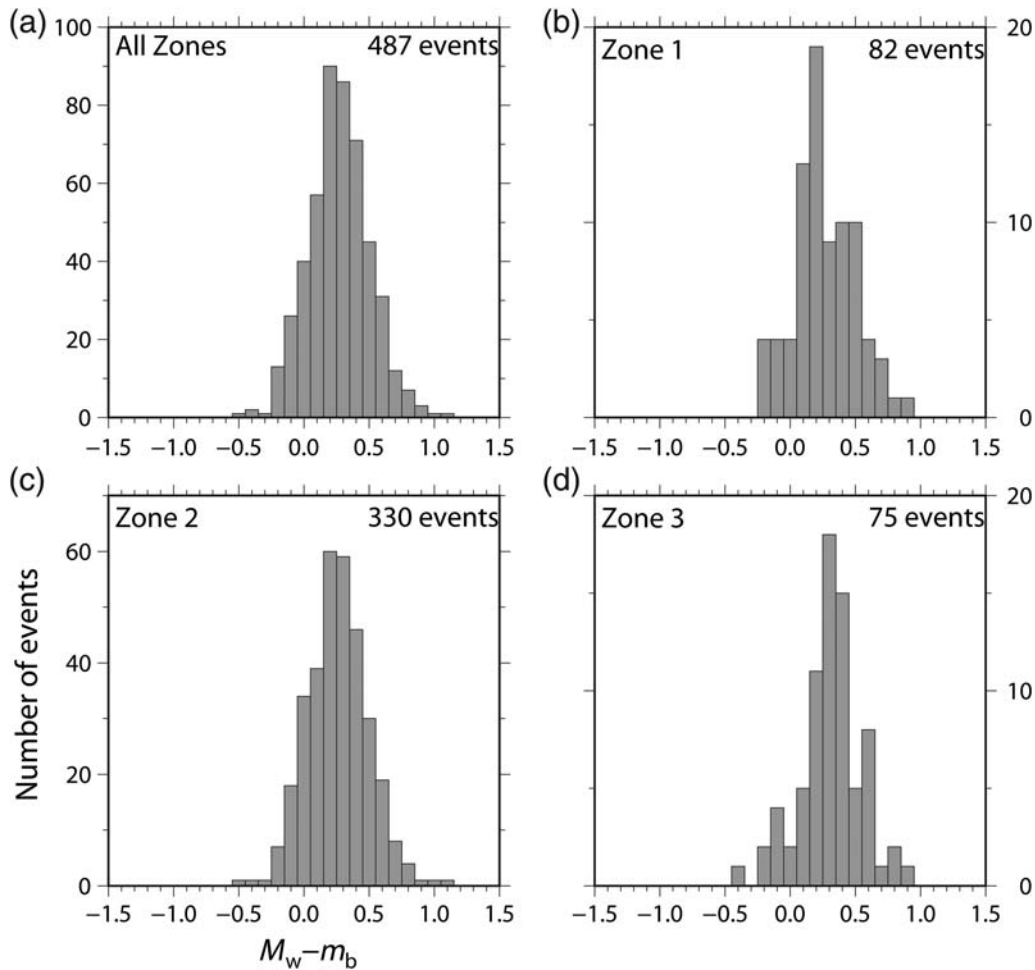


Figure 5. Histograms of the M_w – m_b differences for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. The differences are mainly between -0.2 and 0.7 . Numbers of events are also shown in each figure.

Standard deviations are ~ 0.06 in the GOR, whereas they are 0.13 in the OLS.

Discussions and Conclusions

On the basis of previous studies (Castellaro *et al.*, 2006; Lolli and Gasperini, 2012; Lolli *et al.*, 2014; Gasperini *et al.*, 2015), the GOR should perform better regression in the M_w – M_L conversion. Our results show that the GOR standard deviations are about one-half of the OLS standard deviations (Fig. 4), which suggests that the GOR can indeed provide more robust results for the M_w – M_L conversion in western China. To test if the regression results are significantly affected using data for which M_w are lower than the magnitude completeness (M_c), we calculated the M_c in Global CMT and found it to be 4.9 (Fig. 2). We then performed the GOR regression for 446 events having magnitudes ≥ 4.9 . We found similar results using all 487 events, suggesting that data with magnitudes less than the M_c did not significantly affect the results.

Our M_w – M_L regression results derived from both the GOR and OLS methods are different from results suggested by

some previous studies which showed that M_L exhibited an almost 1:1 correlation with M_w (Ristau *et al.*, 2003; Wu *et al.*, 2005; Goertz-Allmann *et al.*, 2011). Two possibilities might explain the discrepancy. One is that M_L is difficult to be accurately calculated in a region of sparse station distribution (Fig. 1). In western China, the S wave in the continental crust converts to the Lg phase, and no crustal direct S -wave phase can be identified in seismograms beyond 100 km (Kennett, 1989). The Lg phase might have a different attenuation feature from the direct S wave, leading to misestimating the maximum average amplitude in equation (1).

Another possibility is that the $R(\Delta)$ of equation (1) is not suitable in calculating the M_L of events at certain epicentral distances. As previously stated, the structures of zones 1–3 are clearly different, and seismic stations are relatively sparse (Fig. 1). In the regular determination of M_L value in China, $R(\Delta)$ is identical for all zones regardless of structural features when the epicentral distances of earthquakes are larger than 440 km (Table S1). Using the same $R(\Delta)$ might bias the results in under- or overestimating the M_L value (Fig. 4).

The M_w – m_b regression calculated by the GOR and OLS method is clearly different. The GOR slopes are systematic

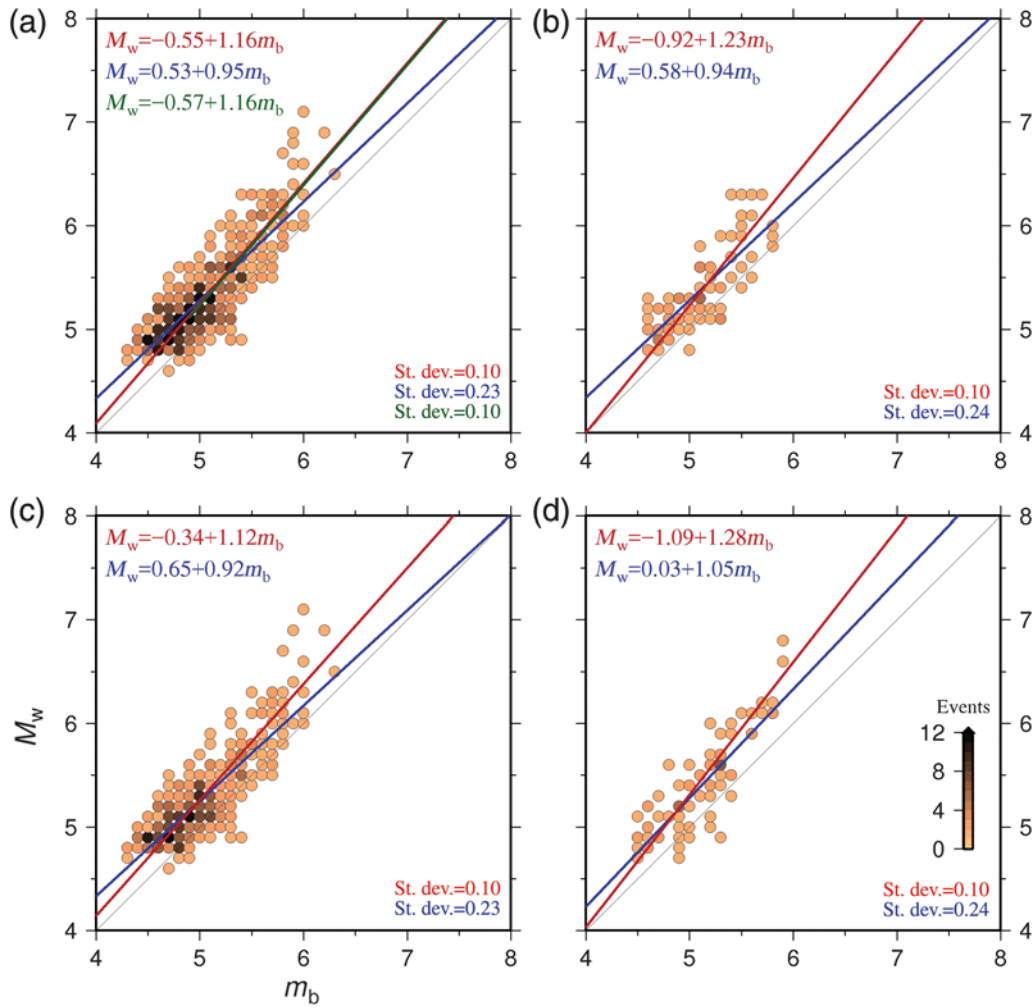


Figure 6. M_w – m_b calibrations for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. Other notations are the same as in Figure 4.

cally ~ 0.2 higher than the OLS slopes for each zone but corresponding to smaller standard deviations (Fig. 6). The result is probably caused due to a combined effect of saturated m_b at ~ 6 and an unsuitable estimation of m_b . In China, m_b is usually estimated using the P wavetrain with a period of ~ 1 s and is calculated using equation (2) for stations whose epicentral distances are between 5° and 100° , regardless of tectonic variances (Table S2). In fact, P -wave attenuations could be significantly distinguished in the three tectonic regions of western China. Our regression results suggest that the $Q(h, \Delta)$ of equation (2) might be modified according to different tectonic units. To clarify this issue, further studies, such as seismic Q values, could help in understanding the attenuation functions (Hearn *et al.*, 2008).

An interesting feature is that the slopes calculated from the GOR method are similar in zones 1 and 3, but those from the OLS are similar in zones 1 and 2. It is possible that the events that occurred in zones 1 and 3 are not enough to reveal the completed M_w – m_b relationship. Alternatively, η of the GOR method is probably not close to 1. In CEDC and Global CMT catalogs, magnitudes are provided without measurement errors, so that we assume that the η equals 1 (Castellaro

et al., 2006). In this study, the investigation of η variation is out of our scope. Further detail investigations about measurement errors in the study regions could help solve this issue.

The M_w – M_s conversion results show relatively small standard deviations for both the GOR and OLS analyses, which indicate that the M_s were consistently calculated by different seismic stations in western China (Fig. 8). However, the slopes of M_w – M_s regression lines deviate from 1, mainly because M_s is larger than M_w for all earthquakes with $M_s > 6.0$. This result should not be simply caused by selection of the constant in equation (3) or distributions of seismic stations. The feature is quite surprising and might indicate a significant misestimate of M_s in western China. The M_s and m_b values are listed in the CEDC catalog together, so that the events used in the M_w – M_s and the M_w – m_b regressions are identical. However, M_w – M_s regression in zone 3 does not show a clear difference from those in zones 1 and 2. It implies that surface-wave attenuations that might cause systematic discrepancy between M_w and M_s are similar in zones 1–3.

Although geological features are clearly different in the three zones, distributions of earthquake sizes are similar (Fig. 2). In zones 1 and 2, moderate-to-large earthquakes

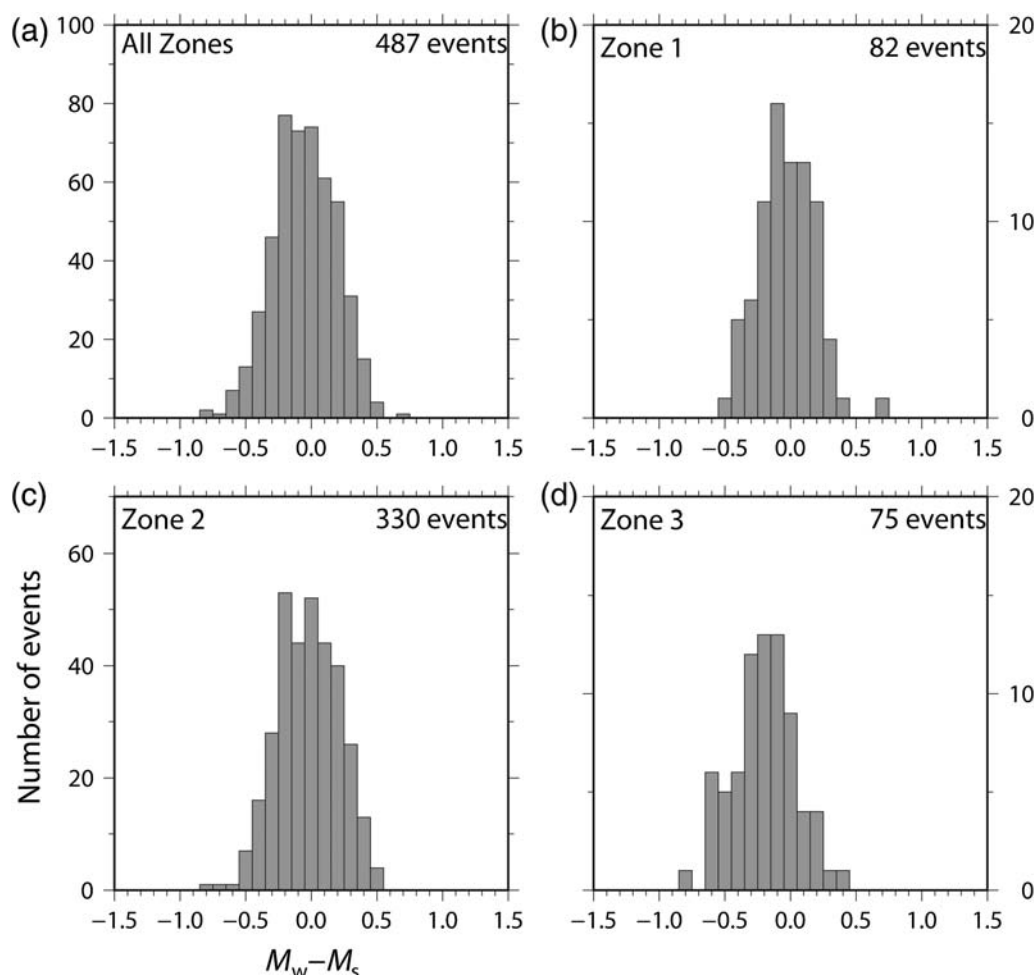


Figure 7. Histograms of the M_w - M_s differences for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. The differences are mainly between -0.4 and 0.4 . Numbers of events are also shown in each figure.

($4.5 \leq M_w \leq 6.0$) are over 95% and 92%, respectively. In zone 3, the percentage of moderate-to-large earthquakes is 88%, which is slightly lower than those in zones 1 and 2. One explanation is that the TP was colliding eastward against the hard south segment of the NSSB (Yang and Liu, 2009; Huang *et al.*, 2013), so rate of stress accumulation was higher than those in the other zones. Such a relatively high rate may result in a larger ratio of large earthquakes ($M_w > 6.0$) to moderate-to-large earthquakes in zone 3. This may be a reason for explaining why the regression slopes of zone 3 are consistently higher than those in other zones. It is also possible that the 25-yr observation is not long enough to show the whole feature of magnitude distributions. Long-term observations in the three zones will help clarify whether or not zone 3 still has the identical scenario in the future.

Lolli *et al.* (2014) used five catalogs (Global CMT, USGS-NEIC moment tensor [MT], European-Mediterranean Regional Centroid Moment Tensor, Eidgenössische Technische Hochschule Zürich Regional MT, and Time Domain Moment Tensor catalogs) to analyze relations only between teleseismic magnitudes (i.e., m_b and M_s) and M_w at global,

Euro-Mediterranean, and Italian scales. They suggested that the relations are nonlinear. Here, we used the Global CMT and the CEDC catalogs to further analyse empirical M_w - M_L , M_w - M_s , and M_w - m_b conversions in western China. We also examined the regression using a global scale conversion provided by Lolli *et al.* (2014). However, standard deviations are ~ 0.26 and ~ 0.14 for M_w - m_b and M_w - M_s conversions, respectively. The standard deviations are higher than those derived from the GOR method, suggesting that the GOR method might be more suitable than the nonlinear regression method.

In this study, we found the standard deviations from the GOR method are clearly smaller than those from the OLS methods. Thus, the GOR method is more suitable for magnitude conversions in western China. Besides, different seismic-wave attenuations might give rise to distinguished line slopes in the three tectonic structures. Some previous studies estimated the relation between different magnitudes, such as M_L - m_b (Gibowicz, 1972; Kiratzi and Papazachos, 1984). We did not calibrate different magnitudes to each other because M_w provides the most robust estimate of the magnitude of large earthquakes. These conversions of magnitude to M_w

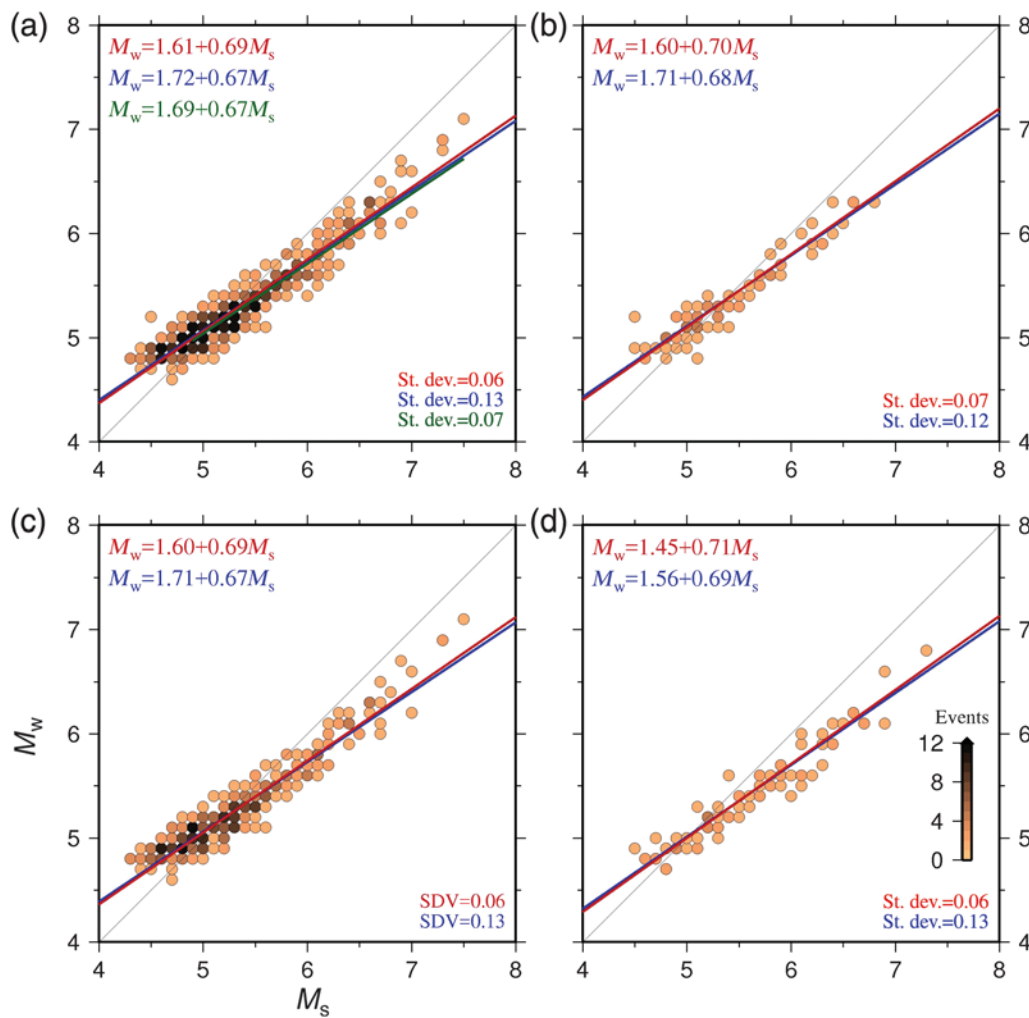


Figure 8. M_w – M_s calibrations for (a) all zones, (b) zone 1, (c) zone 2, and (d) zone 3. Other notations are the same as in Figure 4.

are not only helpful in estimating energy release of a large earthquake, but also allow the western China earthquake database to be particularly useful for mitigation of intensity and seismic hazard after occurrence of a destructive earthquake (Wu *et al.*, 2001). Further, consistency in magnitude estimates of M_w is important for study and comparison of seismicity between different tectonic settings. The conversions will allow for improved seismotectonic studies as well.

Data and Resources

The catalog provided by China Earthquake Data Center was searched using http://www.csndmc.ac.cn/wdc4seis@bj/earthquakes/csn_catalog_p001.jsp (last accessed October 2015; Zheng *et al.*, 2010). Another catalog that consists of M_w produced by the Global Centroid Moment Tensor Project was obtained at <http://www.globalcmt.org/CMTsearch.html> (last accessed October 2015). All figures were made using the Generic Mapping Tools v.4.5.13 (<http://gmt.soest.hawaii.edu/gmt4>, last accessed October 2015; Wessel and Smith, 1998). Topography data were downloaded from the National Geo-

physical Data Center via <https://www.ngdc.noaa.gov> (last accessed June 2015).

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